

Vamana Sesa Sai Siddartha Koppaka

+1 930-333-2483 | siddarthakoppaka@gmail.com | [Linkedin-SiddarthaKoppaka](#) | [Github-SiddarthaKoppaka](#) | [Portfolio](#)

EDUCATION

Indiana University

Master of Science in Computer Science

Bloomington, IN

Aug 2023 - May 2025

Gurukula Kangri (Deemed to be University)

Bachelor of Technology in Computer Science and Engineering

Haridwar, India

Aug 2019 - May 2023

TECHNICAL SKILLS

Programming & Data Processing: Python, SQL, PostgreSQL, PySpark, Apache Spark, Airflow, Kafka, dbt, Snowflake, JS

Machine Learning & AI: Scikit-Learn, TensorFlow, PyTorch, Hugging Face, OpenAI API, LLMs, NLP, Time-Series Analysis, Computer Vision, transformers, LangChain, LangGraph, LangSmith, VertexAI, Ollama, NLTK, Spacy, RAGAS

Cloud & DevOps: AWS (S3, Lambda, ECS, SageMaker), GCP (BigQuery, Cloud Run), Docker, Kubernetes, CI/CD, Redis

Data Visualization: Power BI, Tableau, Matplotlib, Seaborn, Plotly, d3.js

Software Engineering: Large-scale distributed systems, Multithreading, Data processing systems, Pattern Recognition, Natural Language Processing

EXPERIENCE

AI Applications Engineer

July 2025 – Present

Rearc

Remote

- Built RAG systems for clients using LangGraph and LangChain, serving production queries across 50K+ documents with Qdrant vector stores.
- Created End to End AI application evaluation pipelines measuring relevancy, hallucination, faithfulness, and tool call accuracy using DeepEval and custom metrics.
- Implemented guardrails to constrain model outputs and maintain conversation boundaries within client specifications.
- Deployed containerized AI services on AWS and GCP, monitoring inference latency and throughput with Prometheus.

AI Engineer - RA

Nov 2024 – Aug 2025

Kelley School of Business - Indiana University

Bloomington, IN

- Built a RAG system using LangGraph and LangChain, achieving 92% top-3 retrieval accuracy over a Qdrant vector index of 50K+ legal documents.
- Reduced ETL runtime by 20% using PySpark and Dask for batch processing of 10M+ records.
- Fine-tuned LLM models with PyTorch for sentiment classification on 100K+ business reviews, reaching 90% few-shot accuracy.
- Applied NLTK and spaCy for named entity recognition and dependency parsing across 3M+ sentences.
- Deployed Docker containers with Prometheus and Grafana monitoring for inference services on cloud instances.

AI/ML Developer

May 2024 – Aug 2024

Hyphenova

Remote

- Developed features for the Hyphenova application using TypeScript and Next.js, improving response times by 10%.
- Wrote unit and integration tests with Jest and React Testing Library, raising frontend test coverage to 85%.
- Provisioned multi-cloud infrastructure with Terraform on AWS, Azure, and GCP, cutting setup time by 40%.
- Automated CI/CD pipelines using Docker and Terraform, reducing deployment cycles by 40%.
- Built real-time data pipelines with Apache Spark, Kafka, and AWS Data Lake for continuous analytics.

PROJECTS

AI-Powered Document Search Engine | FastAPI, LangChain, Qdrant, Llama 3, Docker, AWS, LLMs

Nov 2024 – Mar 2025

- Built a search system integrating Llama 3 via vLLM with Qdrant VectorDB and LangChain for document retrieval.
- Combined FAISS, BM25, and Qdrant to reduce query latency by 60%, deploying the API on AWS using Bedrock and EC2.

User Churn Prediction & Retention Analysis | Big Data, Databricks, Azure, ML, CI/CD

May 2024 – June 2024

- Processed 50M+ telecom records using Spark on Databricks, automated ETL with Apache Airflow, and stored results in Azure Blob Storage.
- Trained XGBoost and LSTMs, achieving a 92.5% F1-score, and tracked experiments with MLflow.

Thinkwise: AI Idea Evaluator | LangGraph, LangChain, Gemini, FastAPI, MongoDB, ReactJS

Feb 2025 – Apr 2025

- Built a multi-agent system that ranks business ideas based on ROI and effort using Gemini-powered ReAct agents, cutting evaluation time by 70%.
- Handled real-time analysis for 100+ concurrent users with session tracking stored in MongoDB.

Buddy: Your Personal AI Companion | LangGraph, Redis, LangChain, Node.js, TypeScript, Google Cloud

Apr 2025 – Present

- Created a context-aware AI assistant that automates daily tasks, improving productivity tracking by 40%.
- Maintained persistent user sessions across 500+ interactions using Redis for stateful memory and LangGraph.
- Integrated Twilio-based text-to-speech for voice conversations.

CERTIFICATIONS

AWS Cloud Practitioner

Credential ID d8506a288f5b4e4cac01b453f6fece1d

Sep 2024 – Sep 2027

Databricks Academy: Fundamentals Accreditation

Mar 2025 - Mar 2026

Databricks AI Engineer Certification

Dec 2024 - Dec 2027